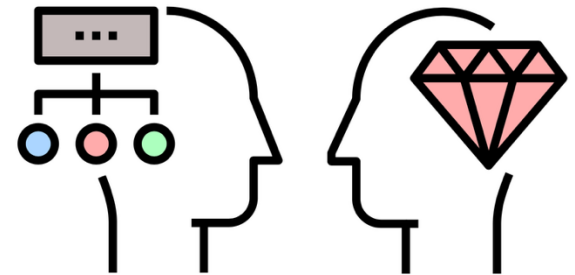


Machine Learning for Materials

2. Machine Learning Basics

Aron Walsh

Department of Materials
Centre for Processable Electronics

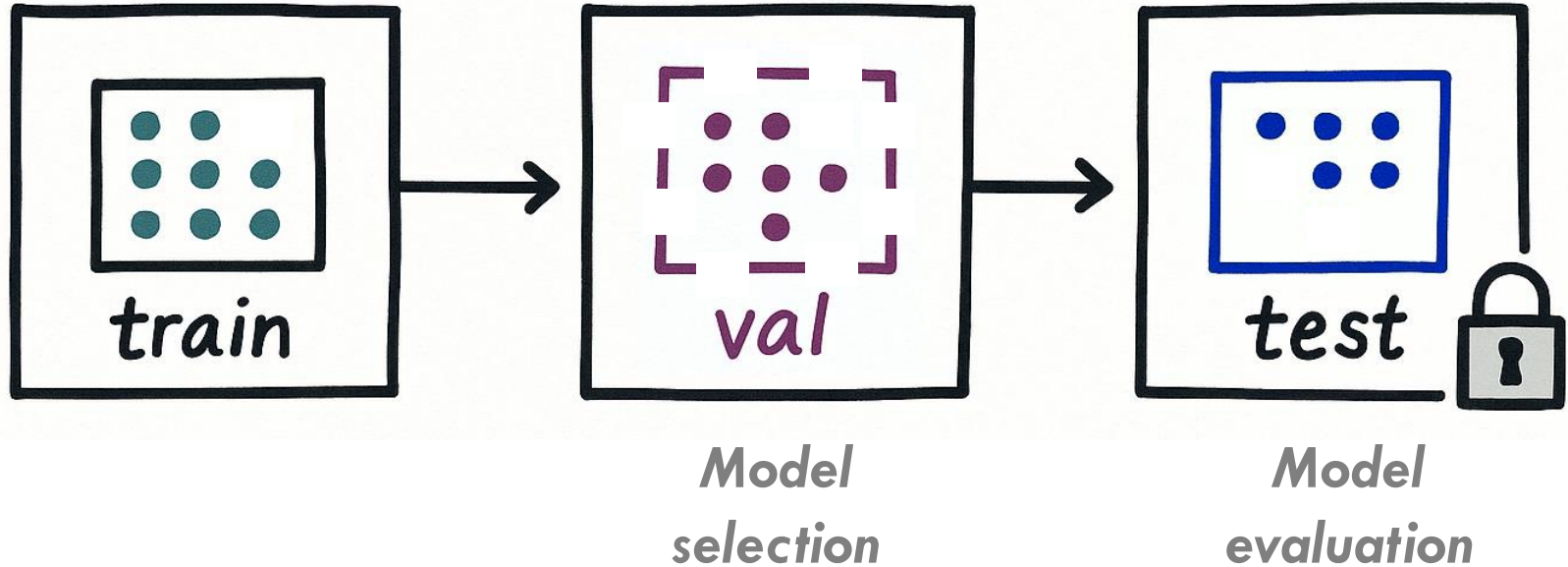


Module Contents

1. Introduction
 - 2. Machine Learning Basics**
 3. Materials Data
 4. Crystal Representations
 5. Classical Learning
 6. Deep Learning
 7. Building a Model from Scratch
 8. Accelerated Discovery
 9. Generative Artificial Intelligence
 10. Future Directions
-

Key Concept #2

Fit → Validate → Generalise



Trust the test set performance, not the training loss

Artificial Intelligence

Computational techniques that mimic human intelligence

ARTIFICIAL INTELLIGENCE (AI)

(entire knowledge field)

MACHINE LEARNING (ML)

(data-driven statistical models)

Supervised

Unsupervised

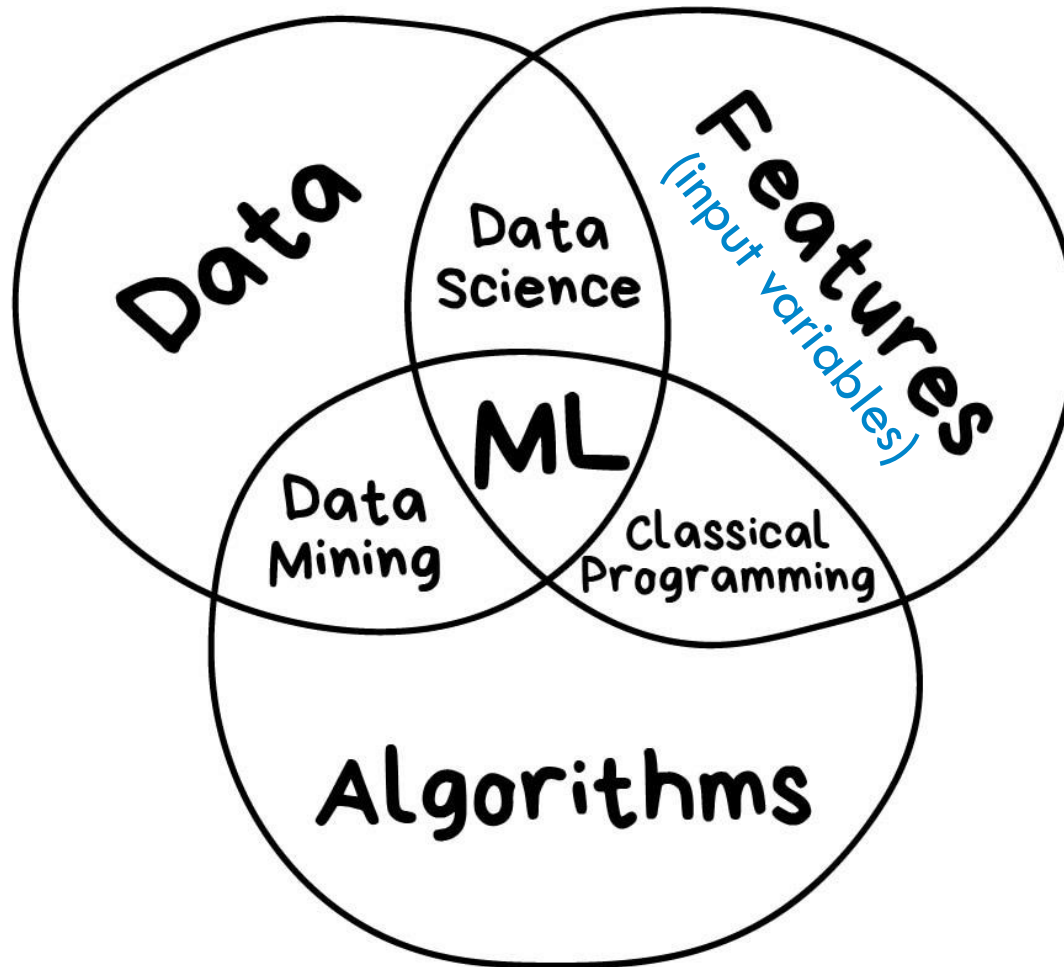
Reinforcement

DEEP LEARNING

(multi-layered neural networks)

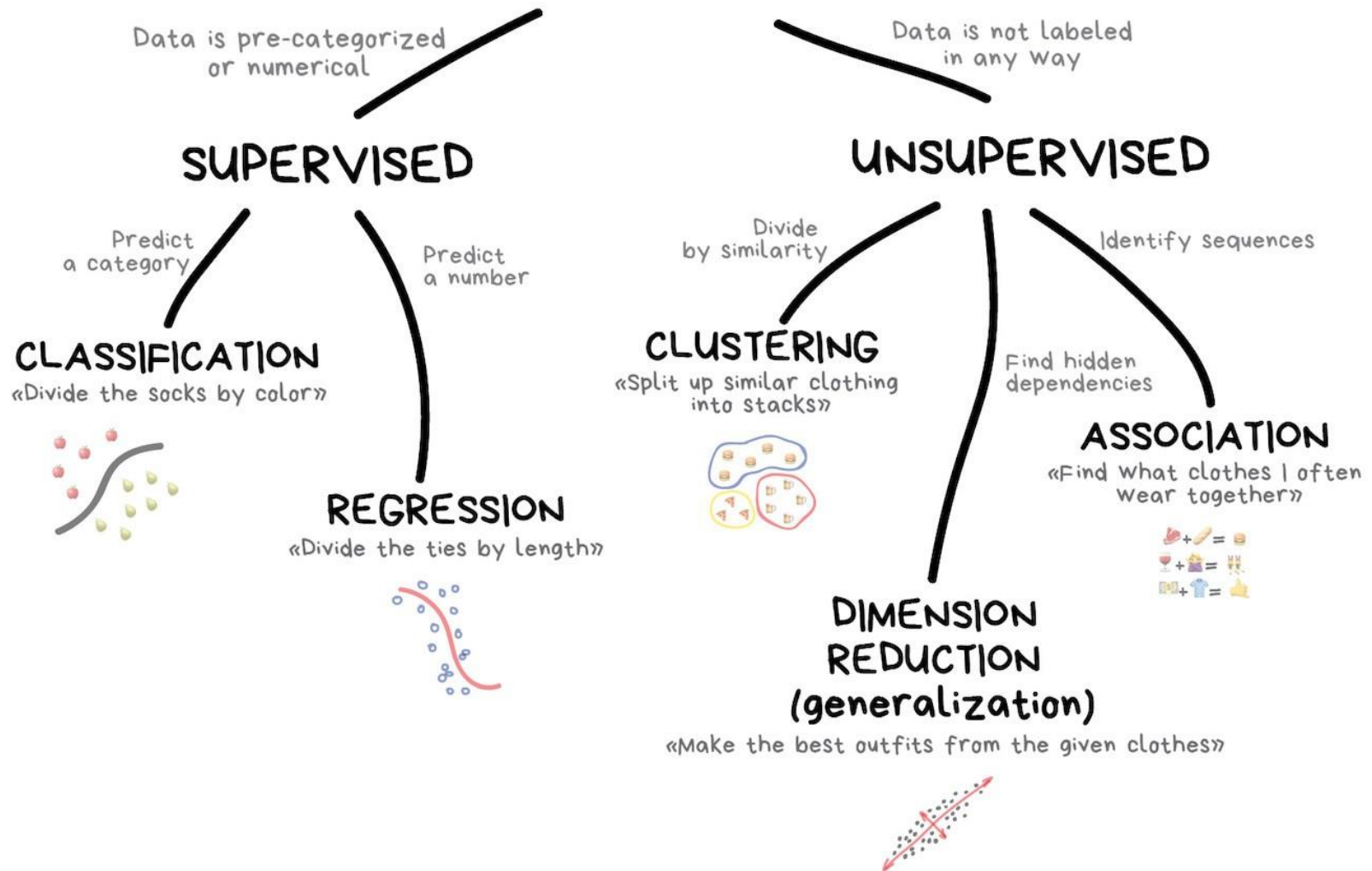
Focus on Machine Learning (ML)

Statistical techniques that improve with experience



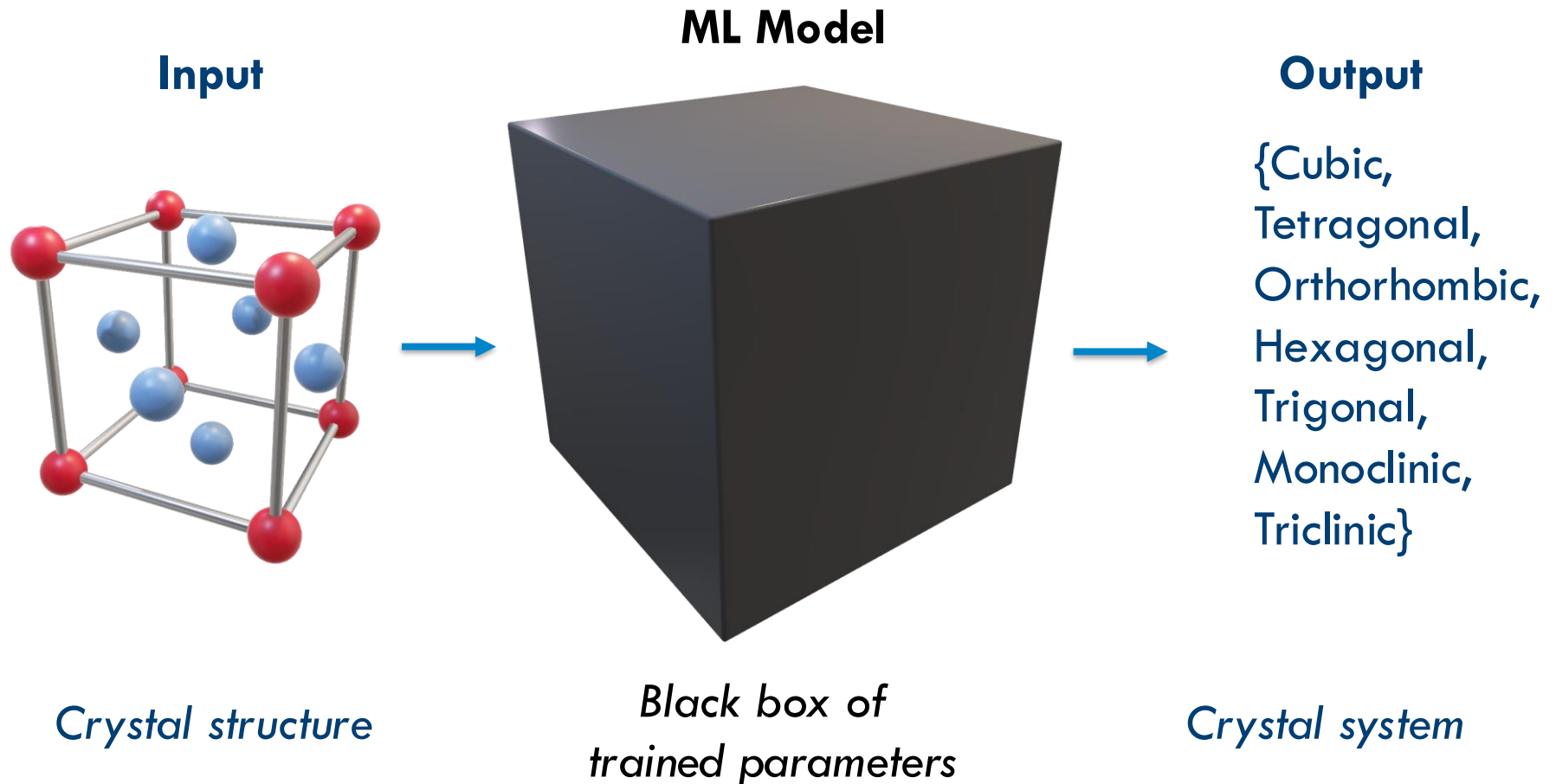
Focus on Machine Learning (ML)

CLASSICAL MACHINE LEARNING



Quiz

What type of learning is this?



“Black box” is a common criticism of ML – with the right tools you can open it up!

Class Outline

Materials Learning Basics

A. Terminology

B. Evaluation metrics

C. Learning by example

Function Approximation

$$\begin{array}{ccc} \text{Output} & & \text{Input} \\ \mathbf{y} & = & \mathbf{f}(\mathbf{x}) \\ \text{Property} & & \text{Features} \\ & & \begin{array}{l} 1. \text{ Composition} \\ 2. \text{ Structure} \end{array} \end{array}$$

Linear Regression

$$\begin{array}{ccccccc} \text{Property} & & \text{Constant} & & \text{Composition} & & \text{Structure} \\ \mathbf{y} & = & \beta_0 & + & \beta_1 \mathbf{x}_1 & + & \beta_2 \mathbf{x}_2 \\ & & \text{Learned} & & & & \\ & & \text{weights} & & & & \end{array}$$

Function Approximation

Generalised Linear Models

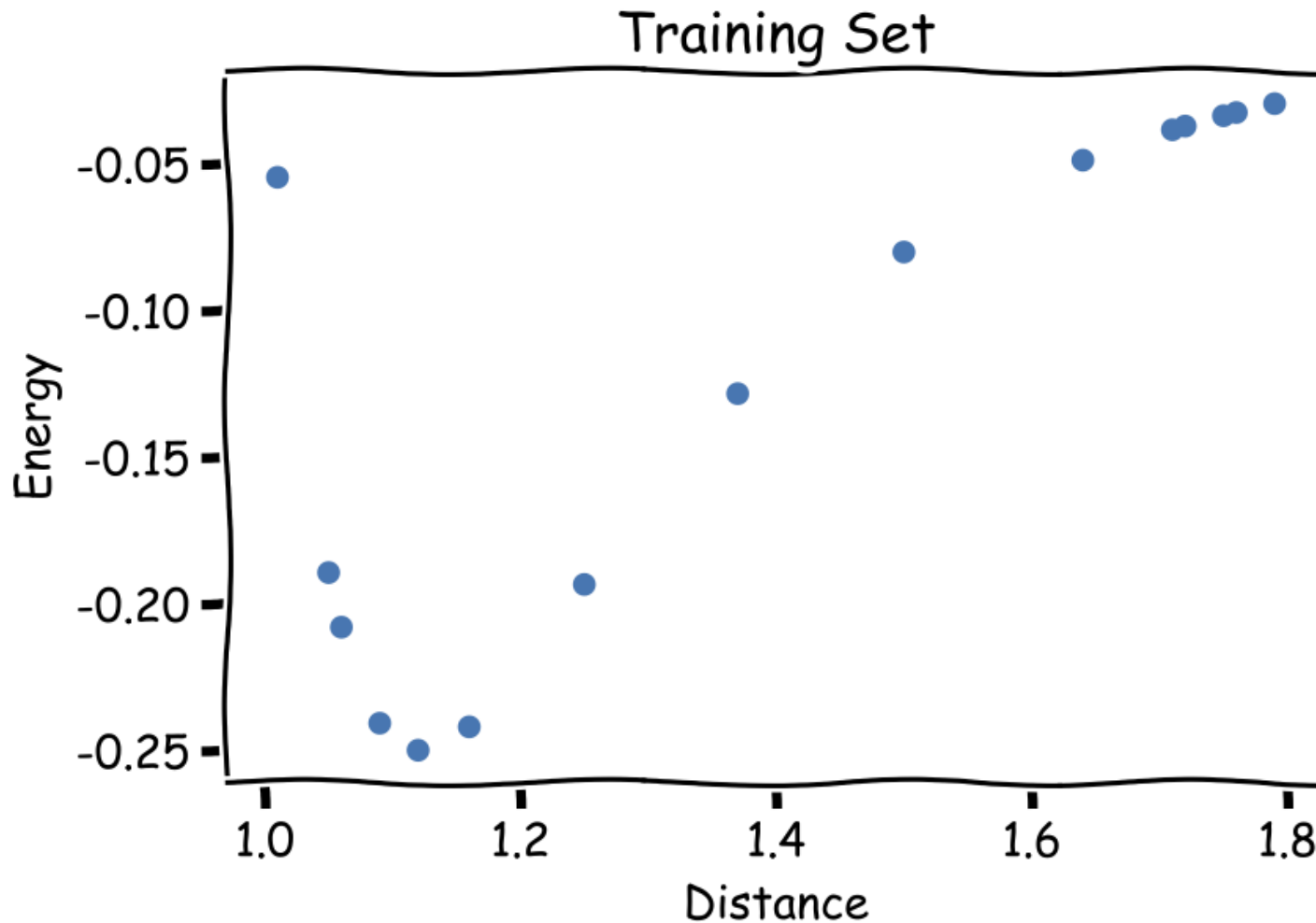
$$\begin{array}{c} \text{Property} \end{array} \quad y = \begin{array}{c} \text{Constant} \end{array} \beta_0 + \begin{array}{c} \text{Composition} \end{array} f_1(x_1) + \begin{array}{c} \text{Structure} \end{array} f_2(x_2)$$

Non-Linear Interactions (Feature Coupling)

$$\begin{array}{c} \text{Property} \end{array} \quad y = \begin{array}{c} \text{Constant} \end{array} \beta_0 + \begin{array}{c} \text{Composition} \end{array} \beta_1 x_1 + \begin{array}{c} \text{Structure} \end{array} \beta_2 x_2 + \beta_3 x_1 x_2$$

Coupling

Function Approximation



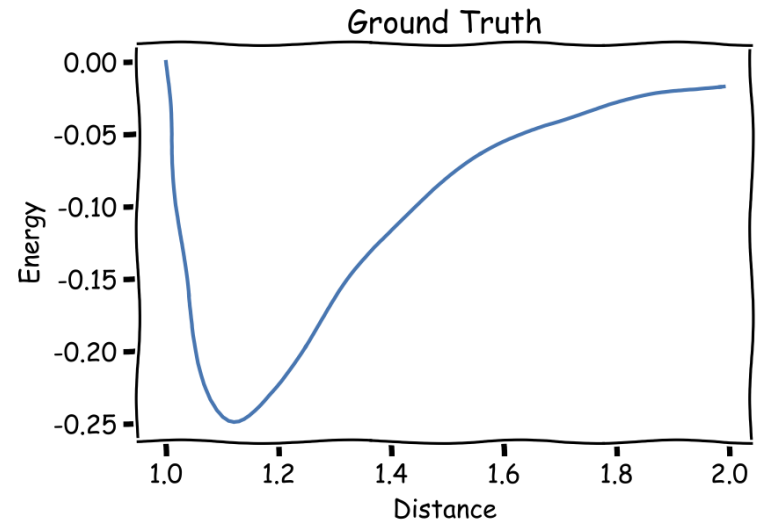
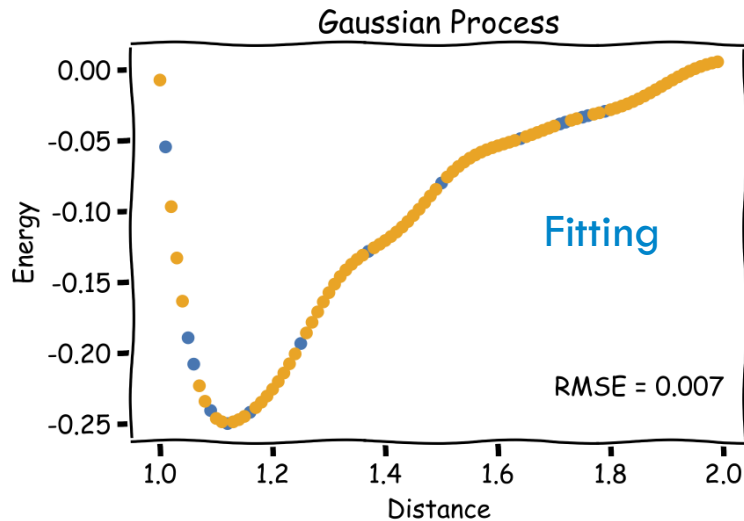
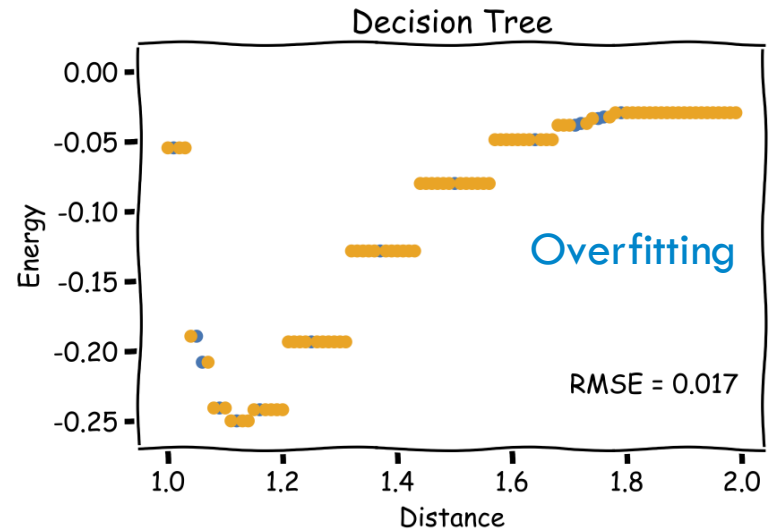
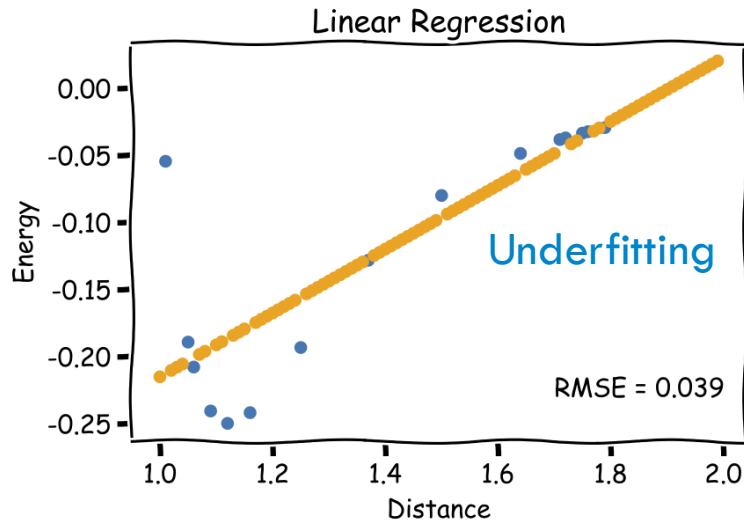
You should recognise the underlying function your undergraduate classes

Function Approximation



```
def truth(r):  
     $\epsilon$  = 1  
    v = ( $\epsilon$ /r)**12 - ( $\epsilon$ /r)**6  
    return v  
  
xvals = np.arange(1, 2, 0.01)  
yvals = truth(xvals)
```

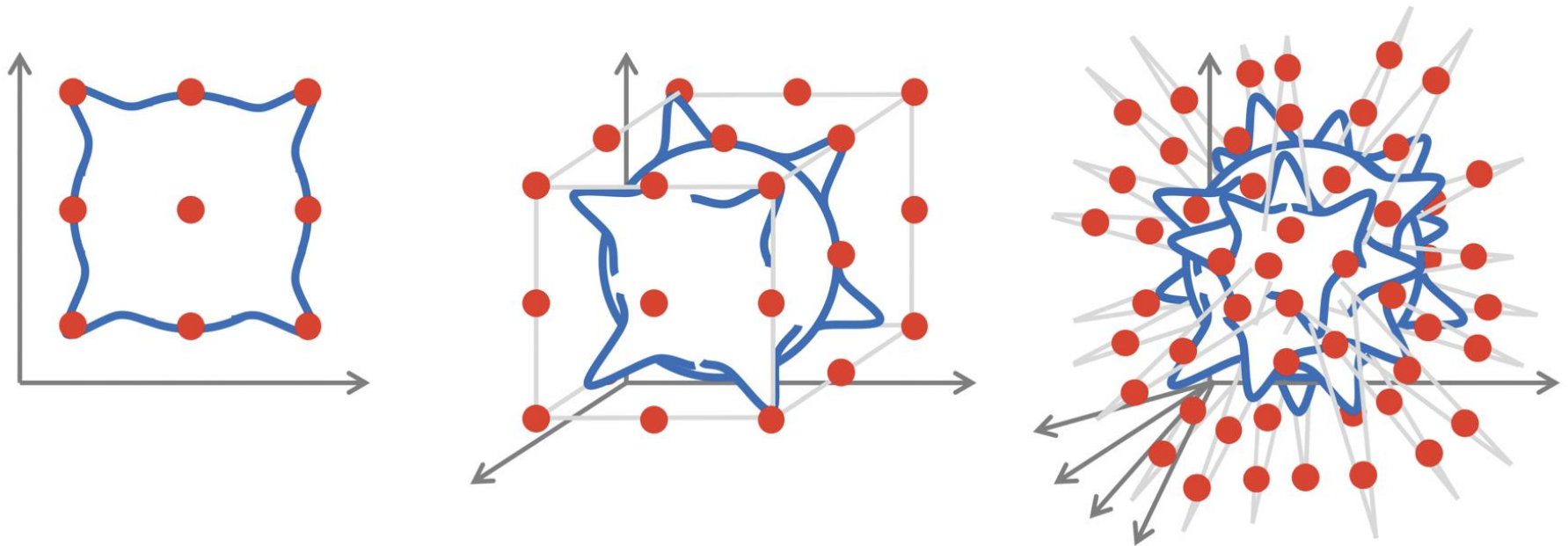
Function Approximation



Function Approximation

Standard expansions work in low dimensions (D).
Real problems face the “**curse of dimensionality**”

An exponential increase in the data requirements
needed to cover the parameter space effectively, $O(e^D)$



Sampling 10 points per dimension: 10^D samples in D dimensions

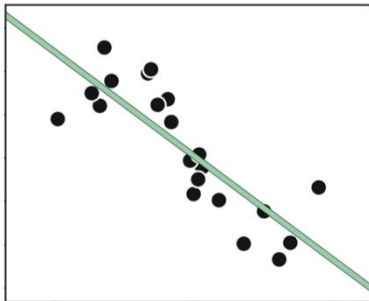
ML Vocabulary

- **Classification model** – input a tensor of *feature values* and output a single discrete value (the *class*)
- **Regression model** – input a tensor of *feature values* and output a continuous (predicted) value
- **Feature** – an input variable
- **Labelled example** – a feature with its corresponding label (the “*answer*” or “*result*”)
- **Ground truth** – reliable reference value(s)
- **Hyperparameter** – model variables that can be tuned to optimise performance, e.g. learning rate

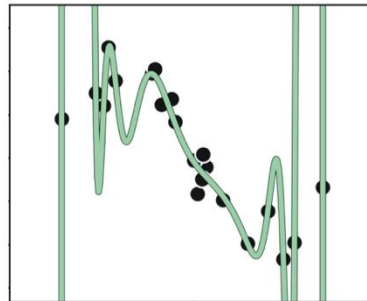
ML Vocabulary

- **Underfitting** – model too simple to describe patterns
- **Overfitting** – model too complex and fits noise

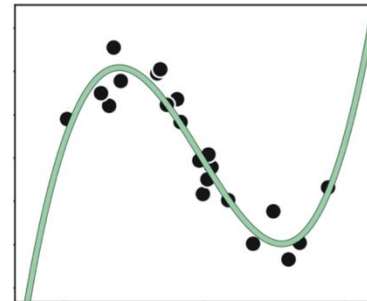
underfitting



overfitting

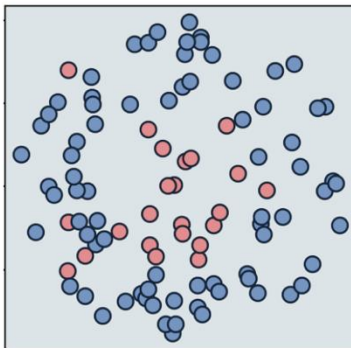


"just right"

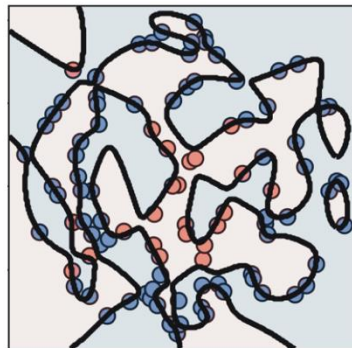


Regression

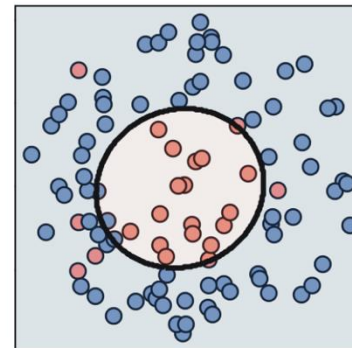
underfitting



overfitting



"just right"



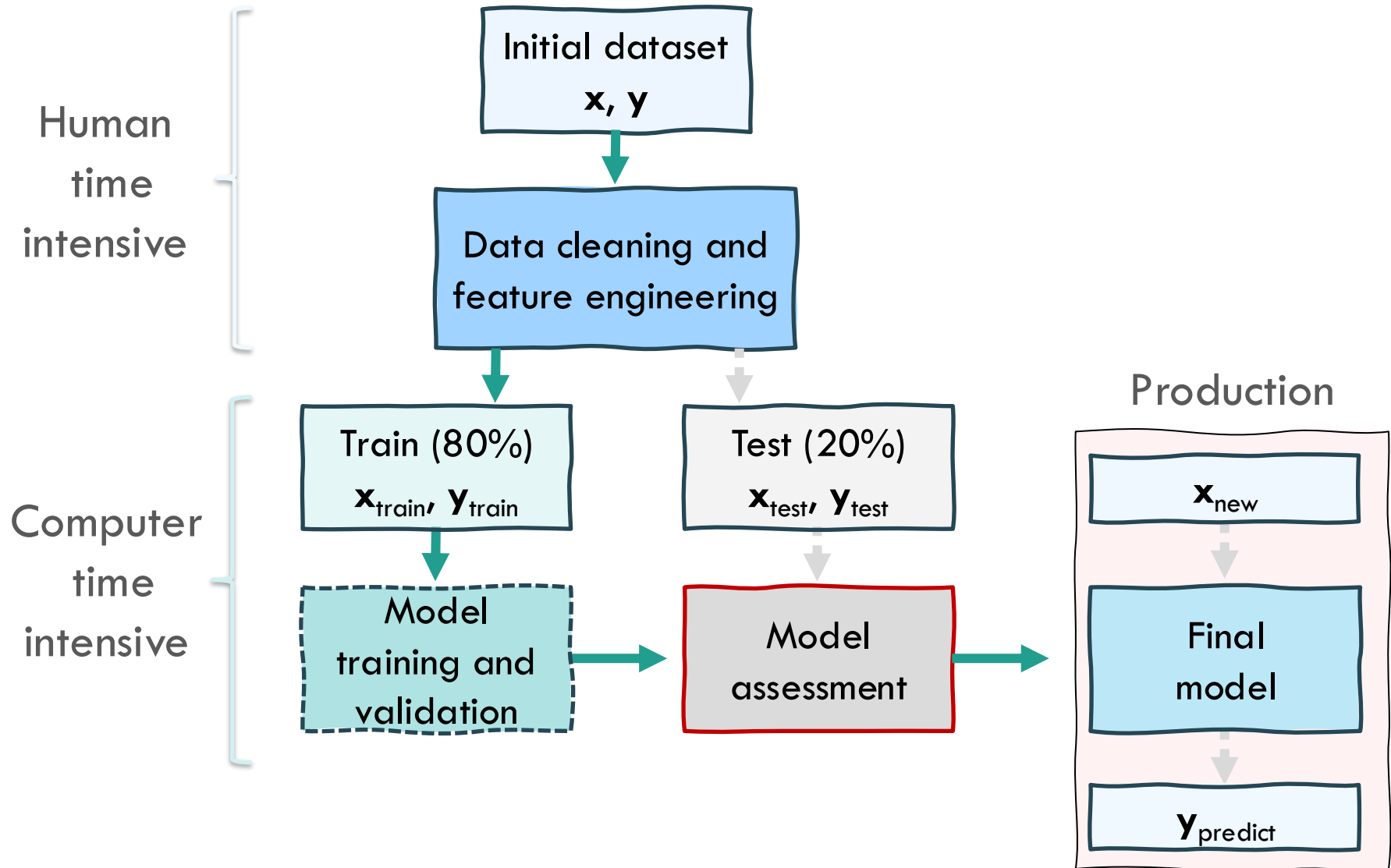
Classification

ML Vocabulary

- **Underfitting** – model too simple to describe patterns
- **Overfitting** – model too complex and fits noise



Typical Supervised ML Workflow



The exact workflow depends on the type of problem and available data

Class Outline

Materials Learning Basics

A. Terminology

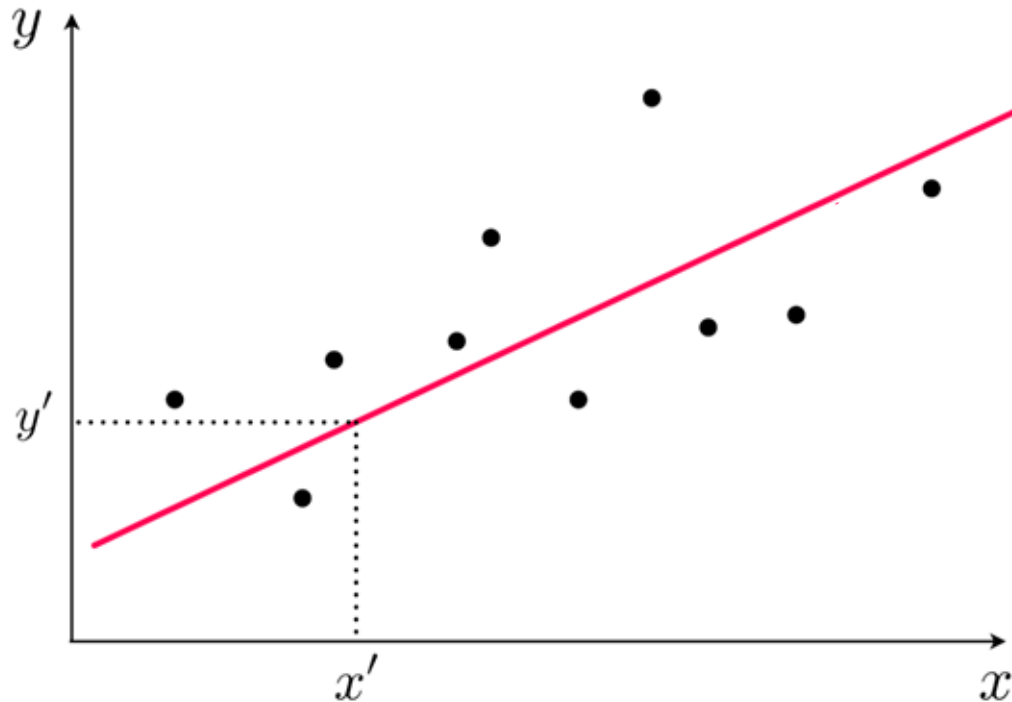
B. Evaluation metrics

C. Learning by example

Model Assessment

Consider a linear model with optimal weights w

$$y^{\text{predicted}} = \text{model}(x, w) = w_0 + w_1 x_i$$



$$\text{Mean squared error (MSE)} = \frac{1}{n} \sum_{i=1}^n (y_i - \text{model}(x_i, w))^2 = \frac{1}{n} \sum_{i=1}^n (e_i)^2$$

Squaring the error ensures non-negativity and penalises larger deviations

Model Assessment

- **Residual** – a measure of prediction error $e_i = y_i - y_i^{\text{predicted}}$
- **MAE** – Mean Absolute Error $= \frac{\sum_{i=1}^n |e_i|}{n}$
- **RMSE** – Root Mean Square Error $= \sqrt{\frac{\sum_{i=1}^n (e_i)^2}{n}}$
- **Standard Deviation** – a measure of the amount of dispersion in a set of values. Small = close to the mean. Expressed in the same units as the data, e.g.

lattice parameters $a = 4 \text{ \AA}, 5 \text{ \AA}, 6 \text{ \AA}$

mean $= (4+5+6)/3 = 5 \text{ \AA}$

deviation $= -1, 0, 1$; deviation squared $= 1, 0, 1$

sample variance $\sigma^2 = (1+0+1)/2 = 1$

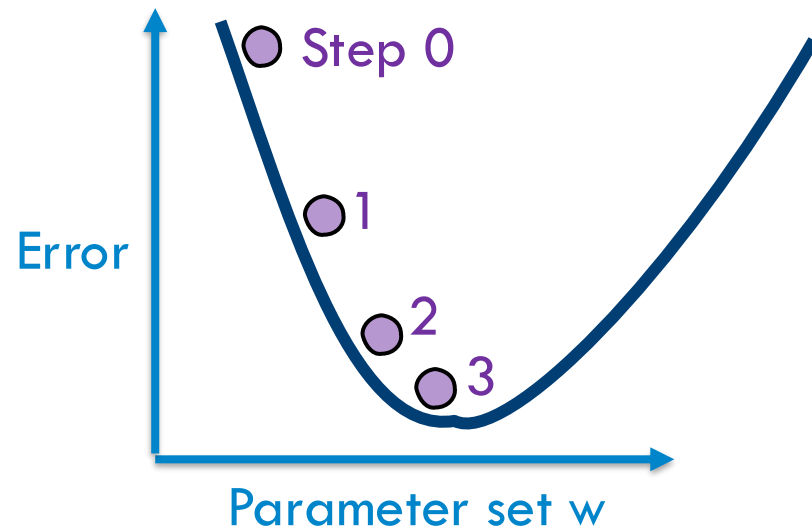
standard deviation $\sigma = 1 \text{ \AA}$

Model Training → Minimising Error

Model weights w are adjusted until a cost function (e.g. RMSE) is minimised

Gradient descent is a popular choice:

$$w_i \rightarrow w_i - \underbrace{\alpha}_{\text{Learning rate}} \frac{d \text{ Error}}{dw_i}$$



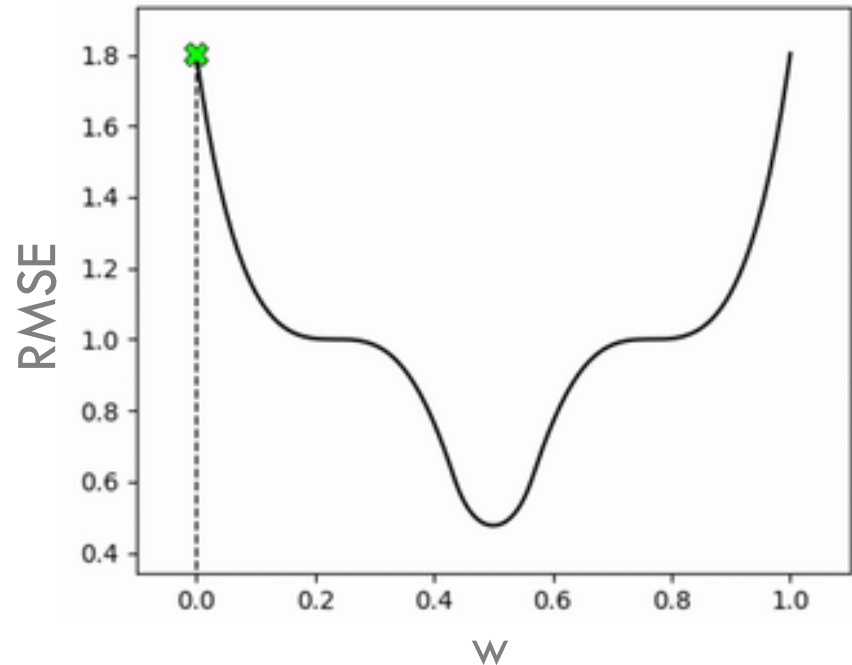
Warning: local optimisation algorithms often miss global minima

Model Training → Minimising Error

Model weights w are adjusted until a cost function (e.g. RMSE) is minimised

Gradient descent
is a popular choice:

$$w_i \rightarrow w_i - \underbrace{\alpha}_{\text{Learning rate}} \frac{d \text{ Error}}{dw_i}$$

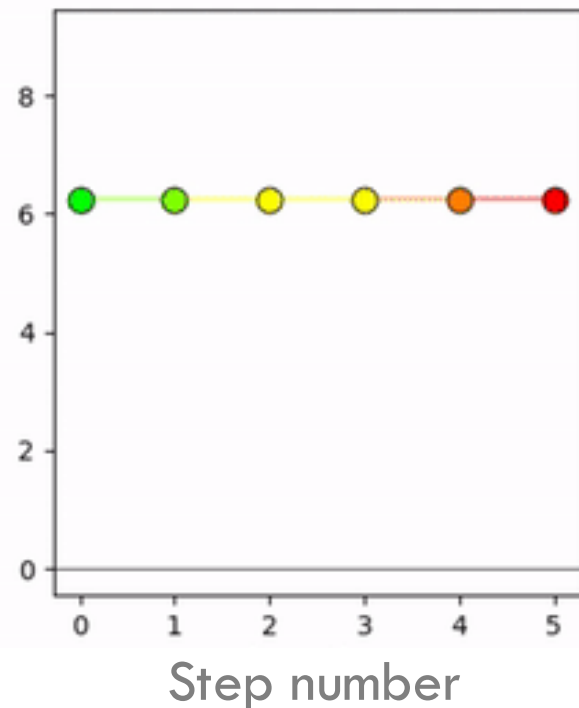
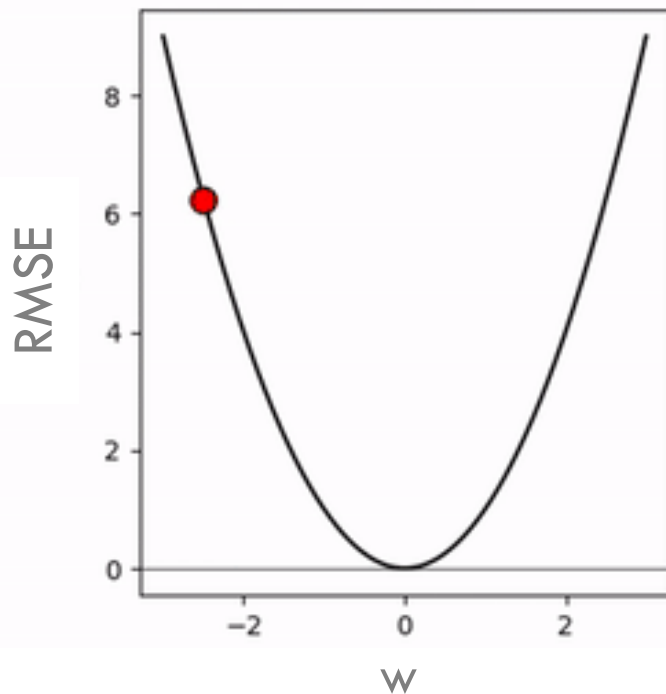


Model Training → Minimising Error

Optimisation algorithms have their own parameters, e.g. step size and no. of iterations

Learning rate (step size)

$\alpha = 0.00$





```
# Define a function and its gradient
def f(x):
    return x**2

def df(x):
    return 2*x

# Initialise starting point and learning rate
x = 5
learning_rate = 0.1

# Perform gradient descent for 10 iterations
for i in range(10):

    # Compute the gradient of the function at x
    grad = df(x)

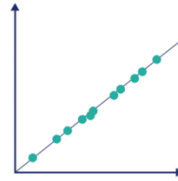
    # Update x using gradient descent
    x -= learning_rate * grad
```

Correlation Coefficient (r)

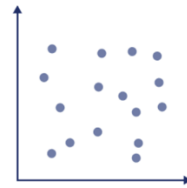
Describes the strength of the relationship between two variables (e.g. “ground truth” vs predicted values)

$$r \in [-1,1]$$

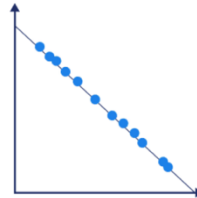
Positive: variables change in the same direction



Zero: no relationship between the variables



Negative: variables change in opposite directions



Pearson correlation*

$$r_{xy} =$$

$$\frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Reminder: correlation does not imply causation

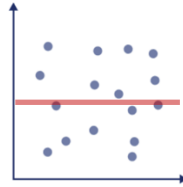
Coefficient of Determination (r^2)

Measure of the goodness of fit for a model.

Describes how well that known data is approximated

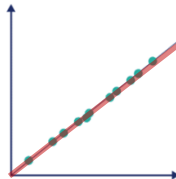
$$r^2 \in [0,1]$$

Zero: baseline model with no variability that predicts $\bar{y}$



0.5: 50% of the variability in y is accounted for

One: model matches observed values of y exactly



Three equivalent definitions

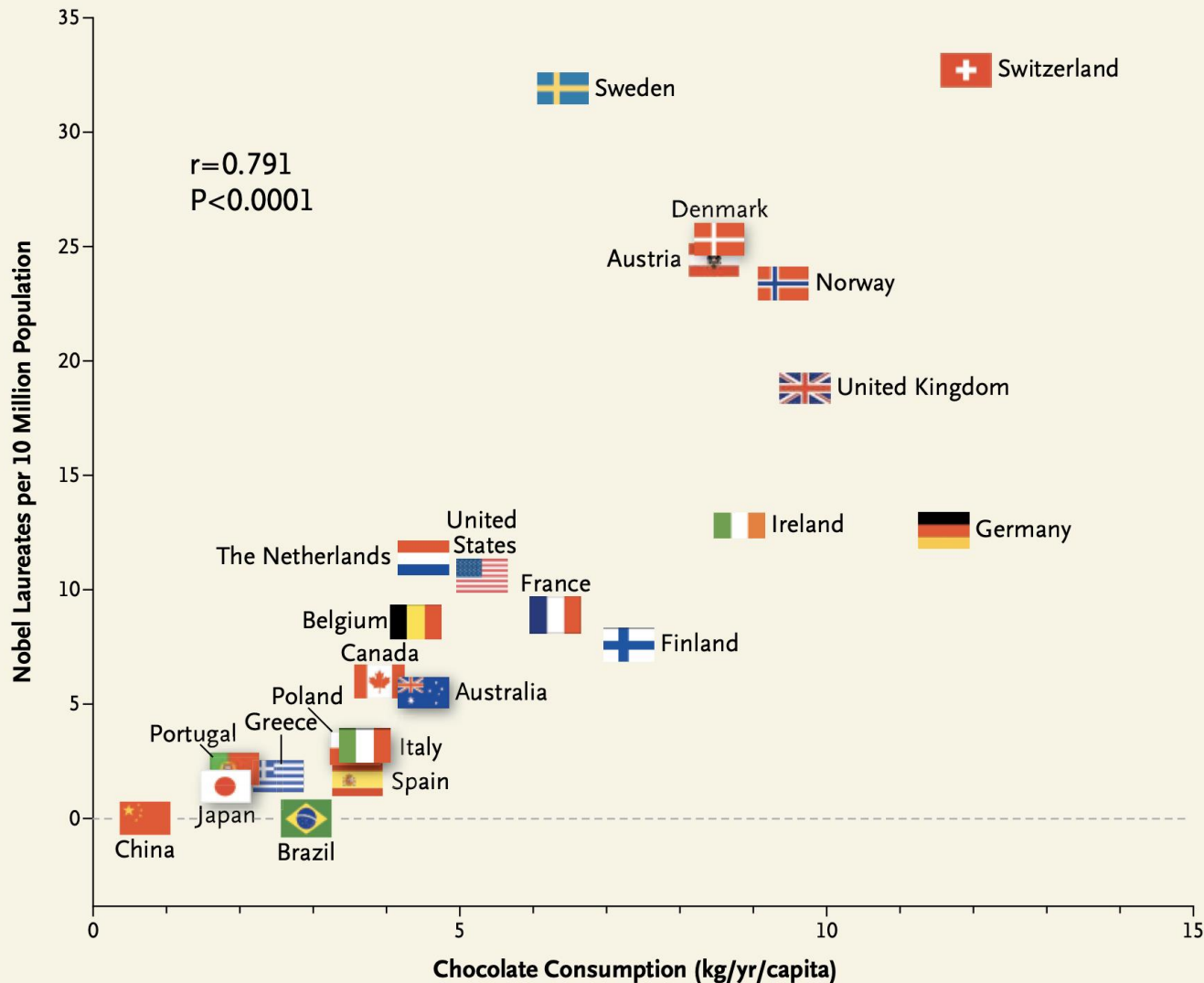
$$r^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

$$r^2 = 1 - \frac{\sum_{i=1}^n (y_i - y_i^{predicted})^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

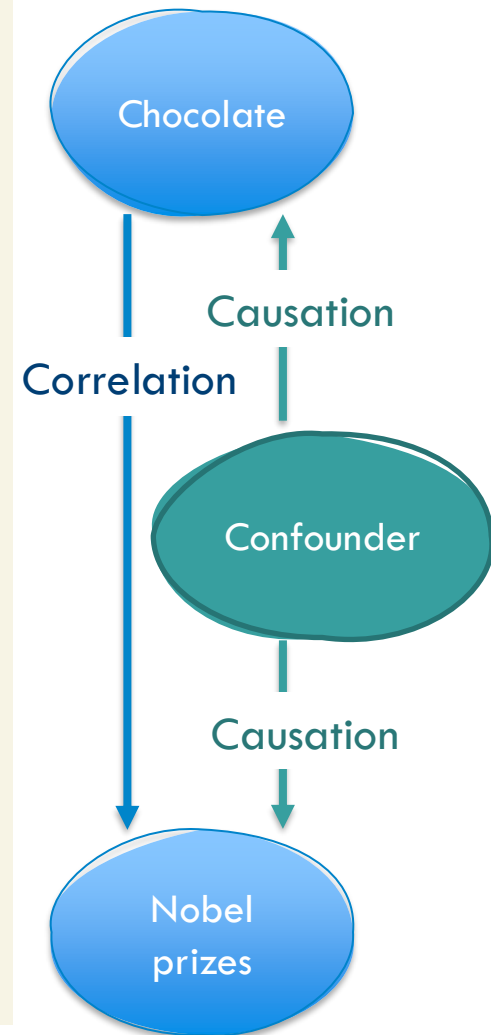
$$r^2 = 1 - \frac{\sum_{i=1}^n (e_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

Note: a unitless metric. Alternative definitions are sometimes used

Correlation, Causation...



Causal Inference



Classification Metrics

Confusion (or error) matrix provides a summary of classification model performance

		Predicted class	
		+	-
Actual class	+	True positive (TP)	False negative (FN)
	-	False positive (FP)	True negative (TN)

$$\begin{bmatrix} 70 & 0 \\ 0 & 30 \end{bmatrix}$$

Perfect model to classify
metals and insulators
($N = 100$)

$$\begin{bmatrix} 66 & 4 \\ 8 & 22 \end{bmatrix}$$

My best
model

Accuracy = Correct/Total

$$(70+30)/100 = 100 \%$$

$$(66+22)/100 = 88 \%$$

Classification Metrics

Confusion (or error) matrix provides a summary of classification model performance

		Predicted class	
		+	-
Actual class	+	True positive (TP)	False negative (FN)
	-	False positive (FP)	True negative (TN)

$$\begin{bmatrix} 70 & 0 \\ 0 & 30 \end{bmatrix}$$

Perfect model to classify
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My best
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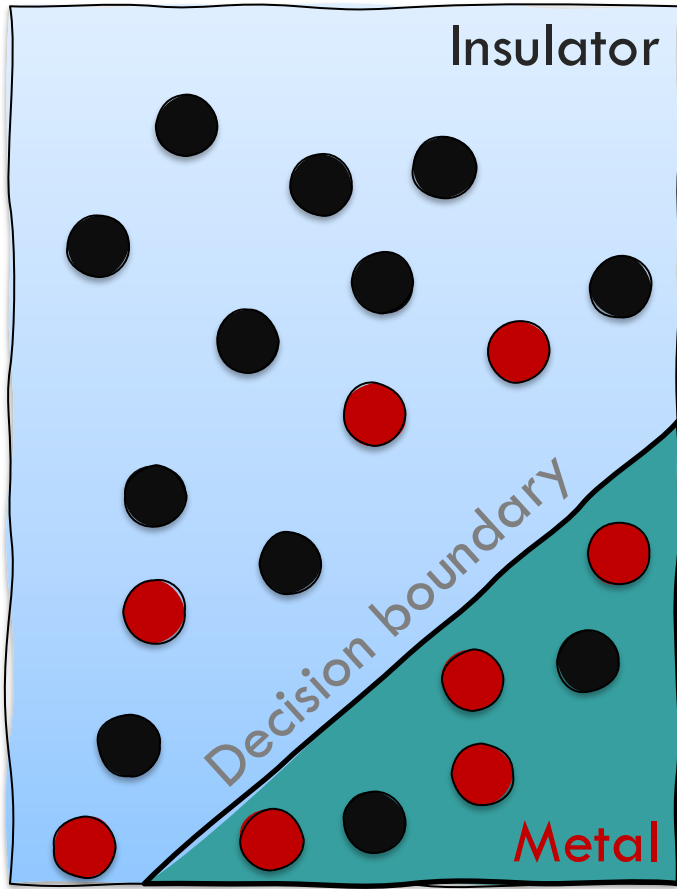
$$\text{Sensitivity} = TP / (TP + FN)$$

$$70 / (70 + 0) = 100 \%$$

$$66 / (66 + 4) = 94 \%$$

Quiz

Fill in “?” for this confusion matrix



		Predicted class	
		Insulator	Metal
Actual class	Insulator	10	2
	Metal	?	4

There are 20 data points in total

Class Outline

Materials Learning Basics

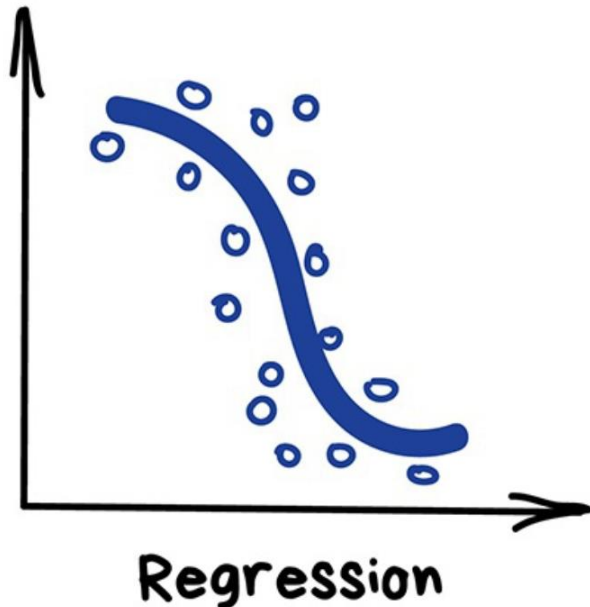
A. Terminology

B. Evaluation metrics

C. Learning by example

Supervised Regression

Model that maps an input to an output based on example input-output pairs (labelled data)



Regression predicts a continuous value, e.g. a reaction rate

Target
variable

Error

$$y = f(x) + \epsilon$$

Learned
function

Regression Example

Predict the dielectric constant of a crystal

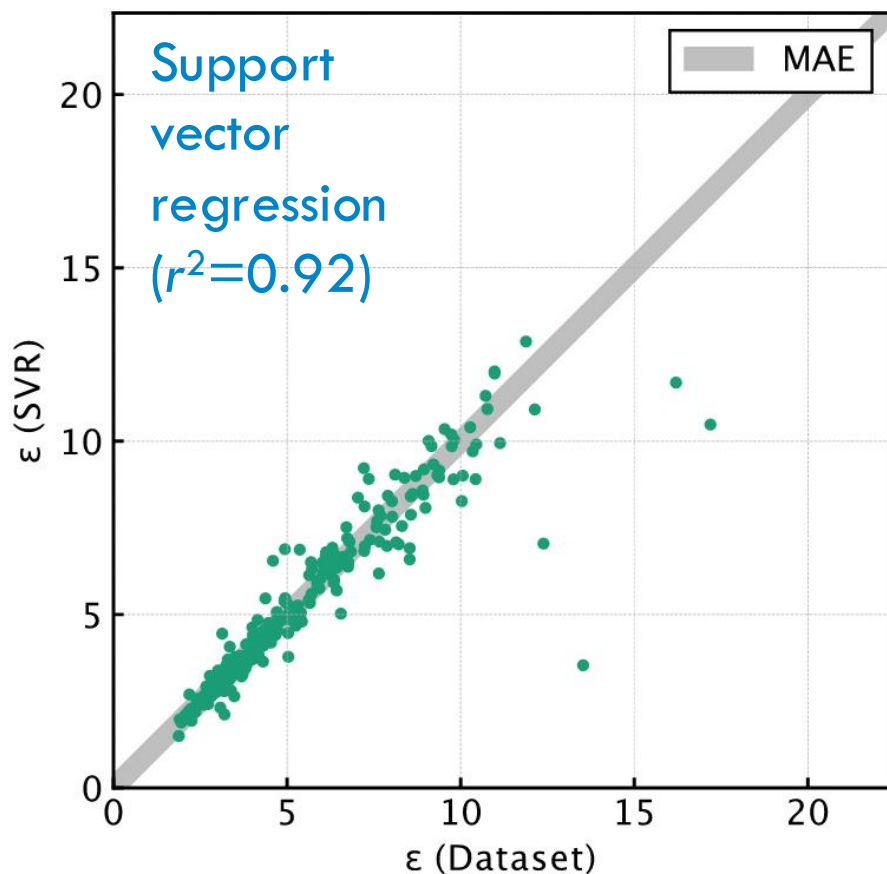


TABLE I. Features used to train the support vector regression model.

Feature	Dimensions
Bandgap ^a	1
Δ Pauling energy ^b	1
Material density ^a	1
Formation energy (per atom) ^a	1
Oxidation state (minimum, variation) ^a	2
Madelung energy (minimum, maximum) ^c	2
Ionic species (one hot encoded) ^c	85

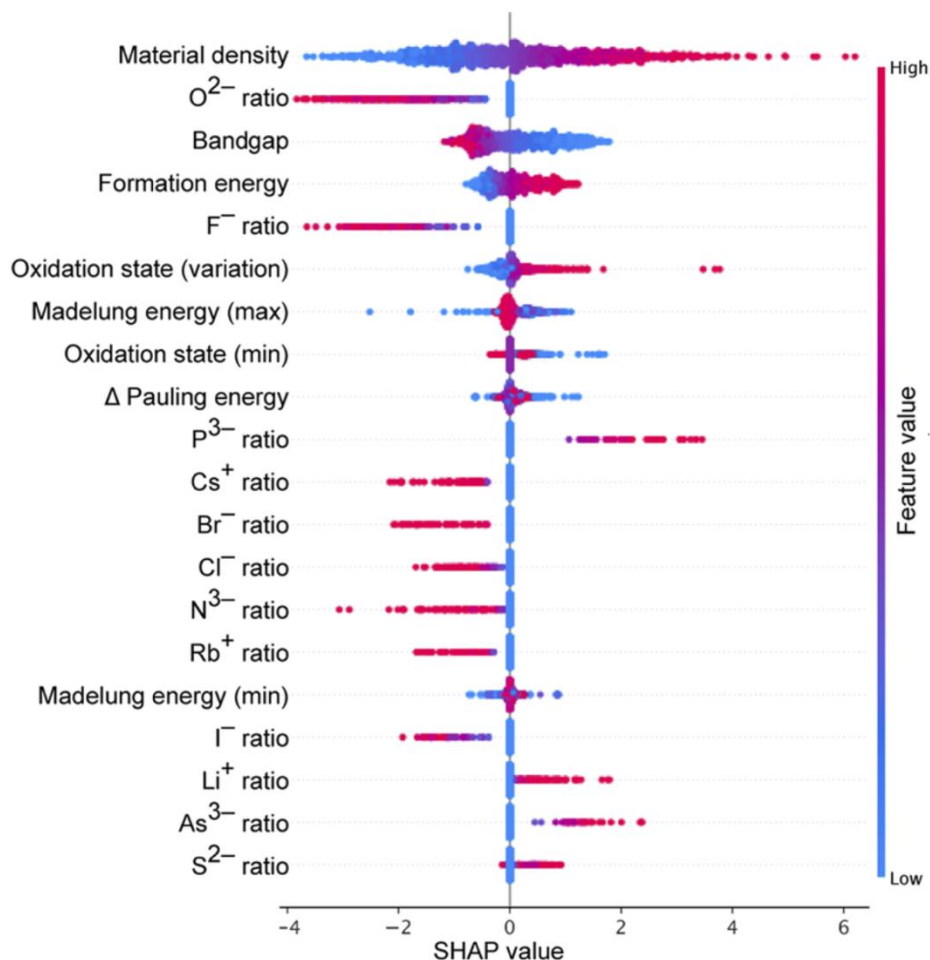
TABLE II. Performance metrics of support vector regression (SVR) and deep neural network (DNN) for training and test data. Metrics are mean Pearson's correlation coefficient (r^2), average error (MAE), mean squared error (MSE), and root mean square error (RMSE).

Metric	SVR		DNN	
	Training	Test	Training	Test
r^2	0.92	0.86	0.95	0.84
MAE	0.24	0.44	0.20	0.55
MSE	0.69	0.99	0.38	1.17
RMSE	0.83	0.99	0.62	1.08

Note: outliers are often interesting cases (poor or exceptional data)

Regression Example

Predict the dielectric constant of a crystal



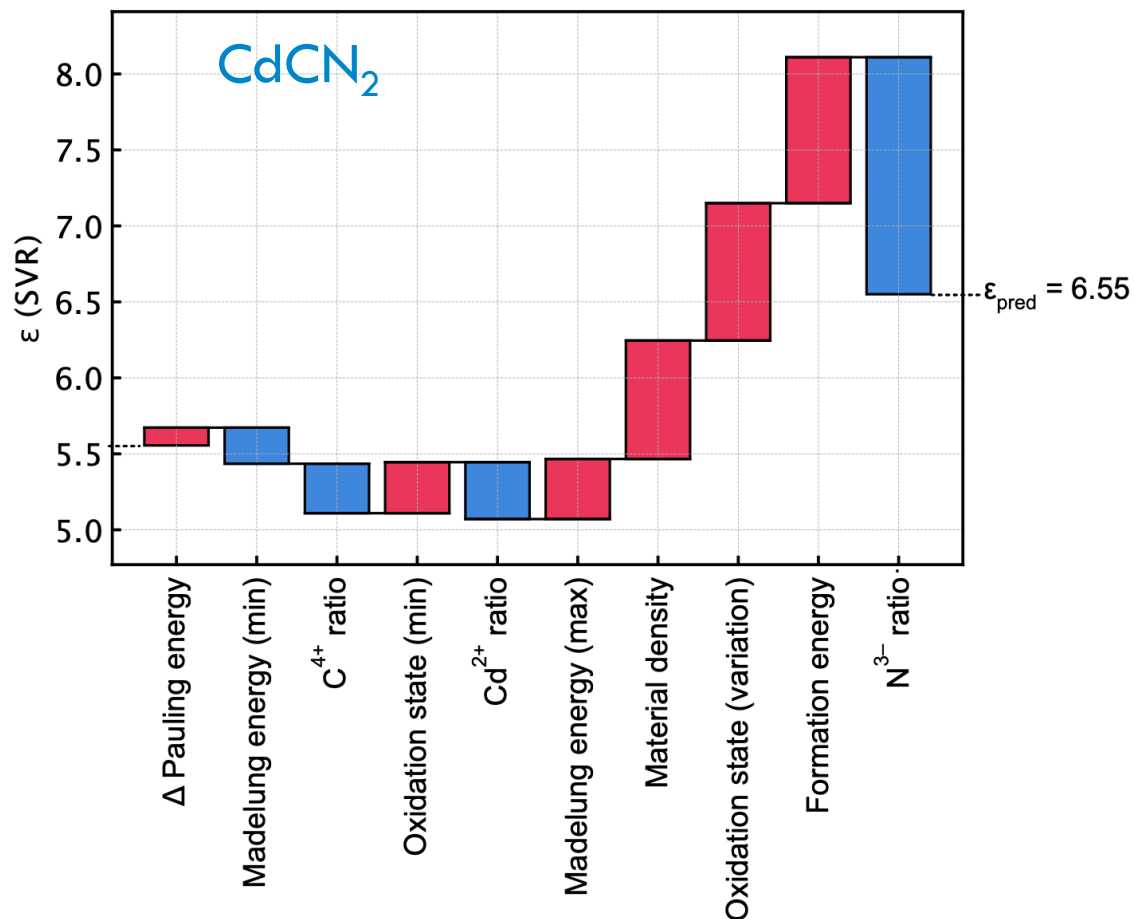
SHAP (SHapley Additive exPlanations) analysis is a method for interpreting ML models

Relative importance of input features in making predictions

A positive SHAP indicates a feature contributes to an increase in the prediction

Regression Example

Predict the dielectric constant of a crystal

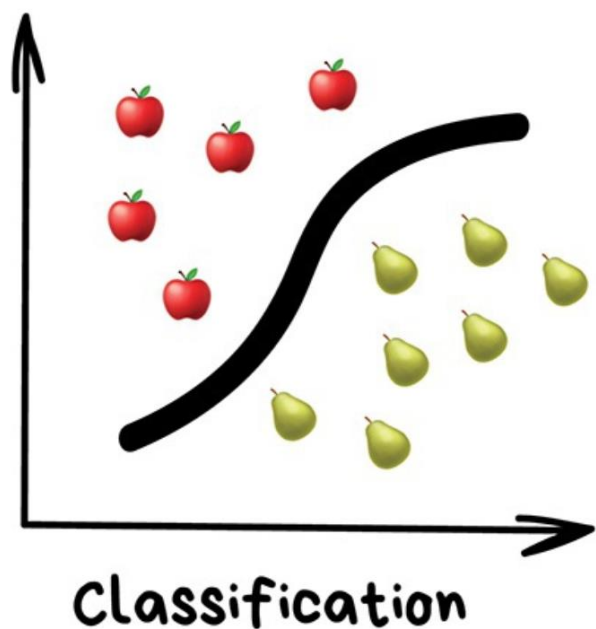


Transparent
breakdown of a
predicted value

*Note: a physical
connection is not implied
(only a correlation)*

Supervised Classification

Model that maps an input to an output based on example input-output pairs (labelled data)



Classification predicts a category, e.g. decision trees for reaction outcomes

Class
label

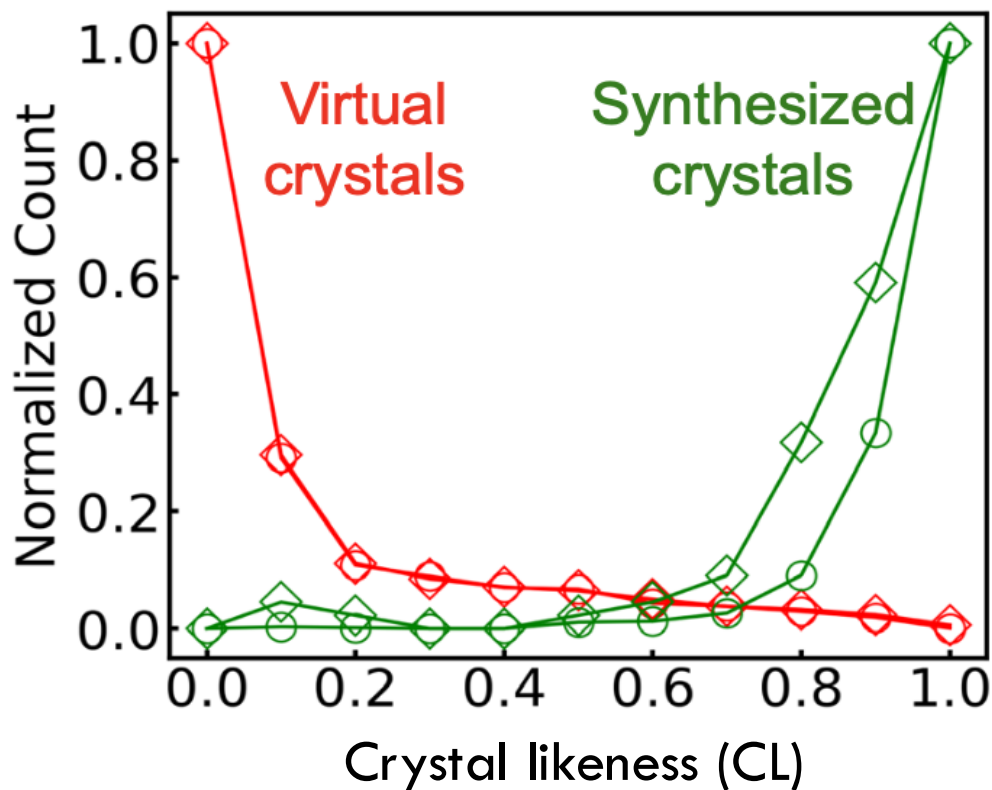
$$y = f(x)$$

Classifier

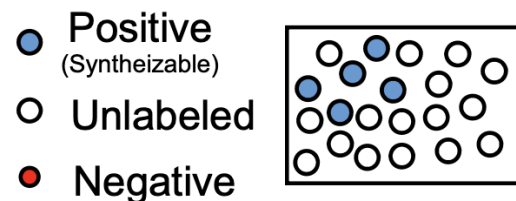
Assignment can be absolute or probabilistic (e.g. 90% apple, 10% pear)

Classification Example

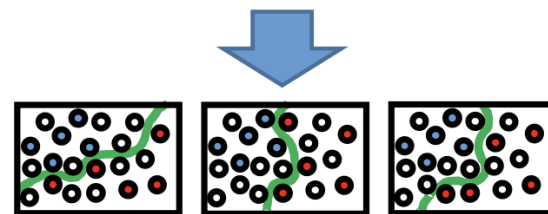
Predict if a material will be stable or unstable



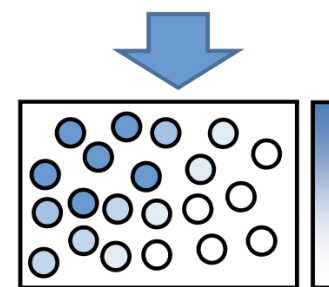
Probabilistic score for the class label



Positive Unlabeled Data



Positive Unlabeled Learning



Synthesizability

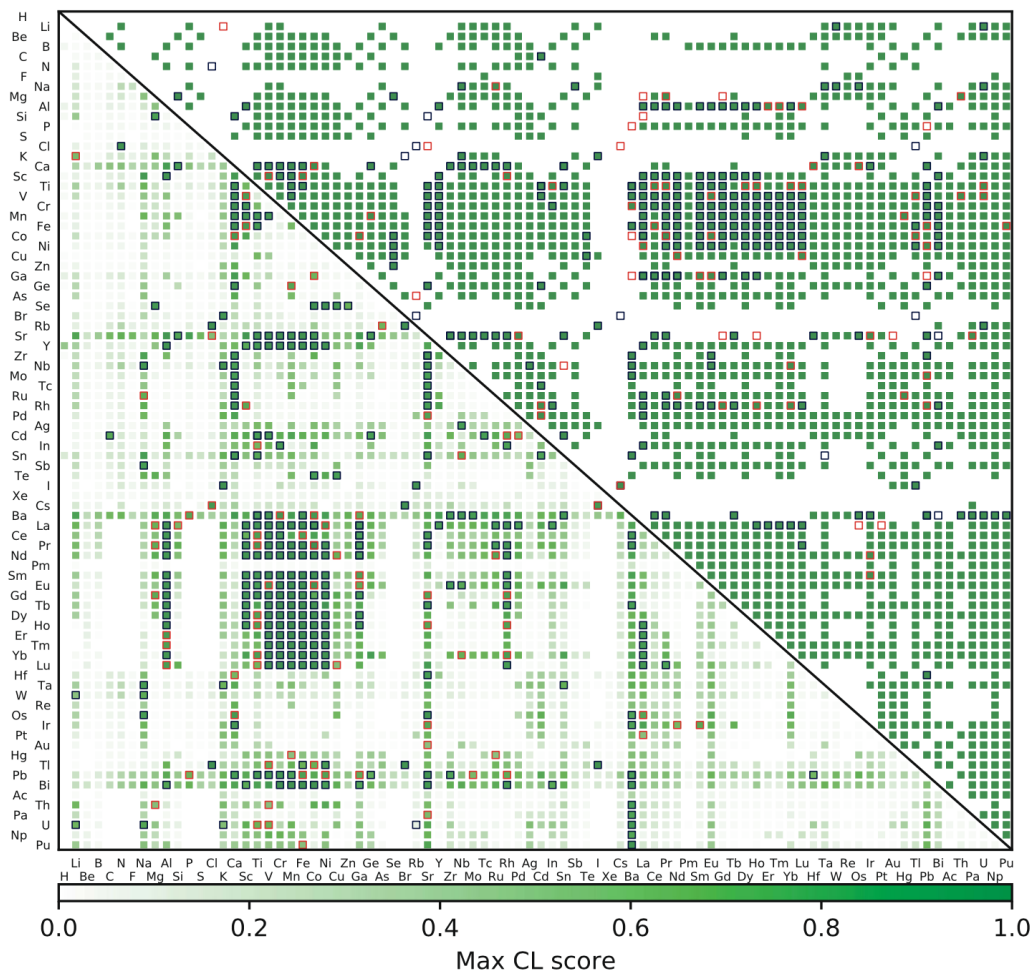
Classification Example

ABX_3 perovskite crystals

Neural
network
model

Improved
selectivity
for promising
compositions

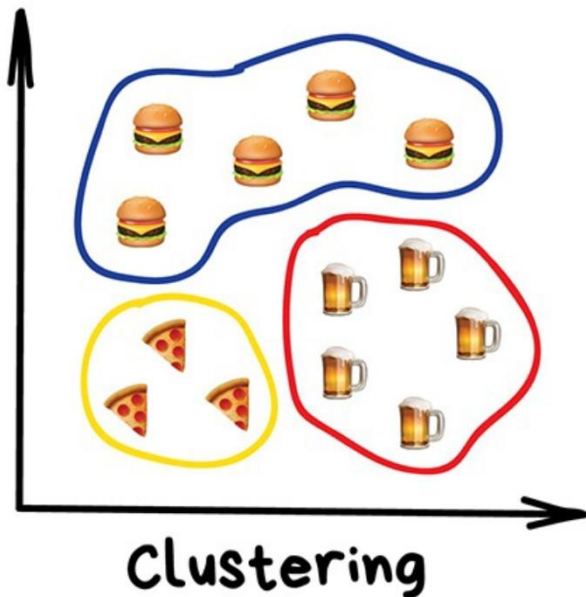
Radius
ratio rules



Likelihood of formation

Unsupervised Learning

Model that can identify trends or correlations within a dataset (unlabeled data)



Clustering groups data by similarity, e.g. high-throughput crystallography

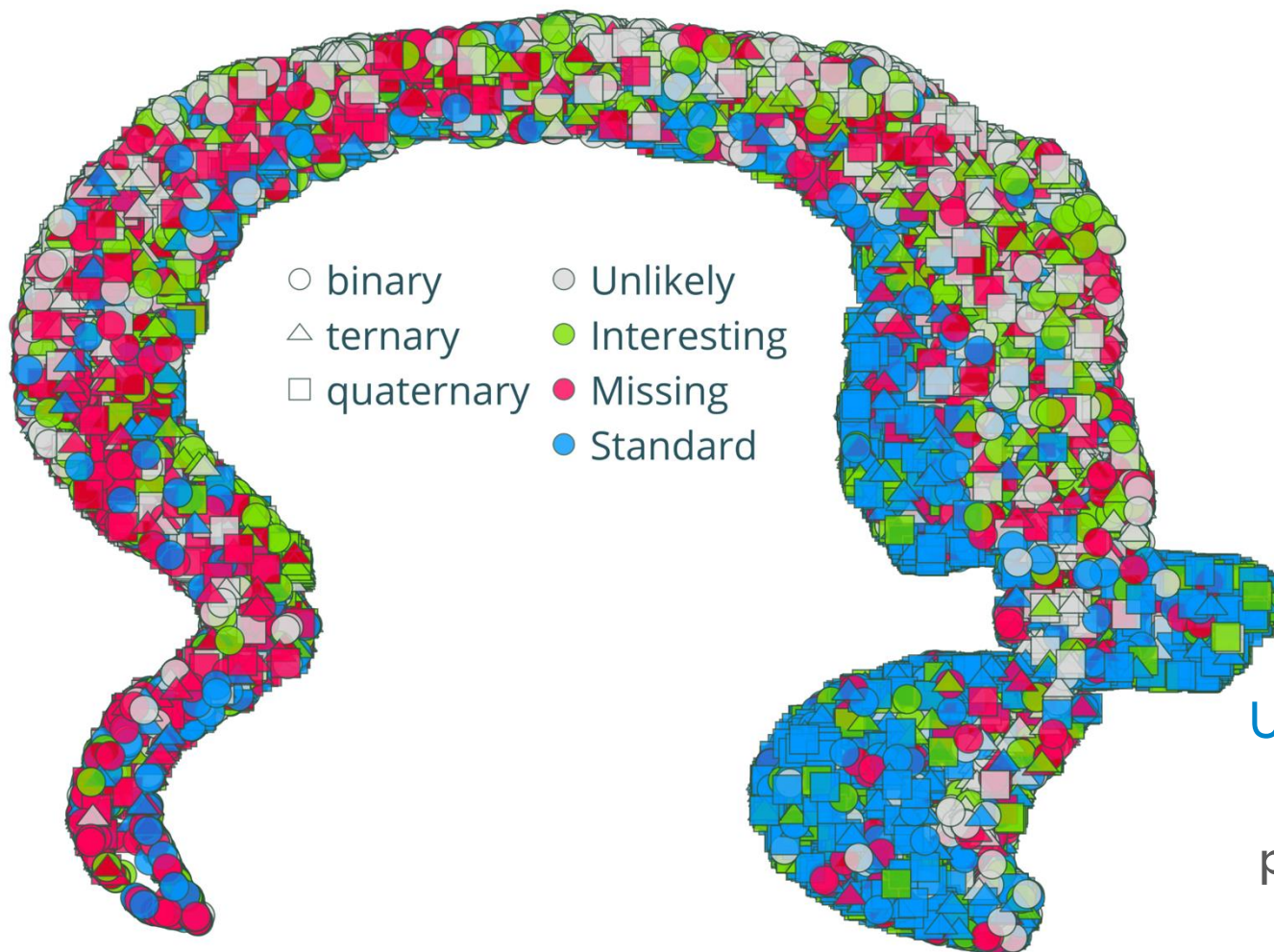
Input data

$$\mathbf{x} \rightarrow f(\mathbf{x})$$

Transformation
to new representation

Unsupervised Example

Map materials space according to their features



Dimensionality
reduction techniques:

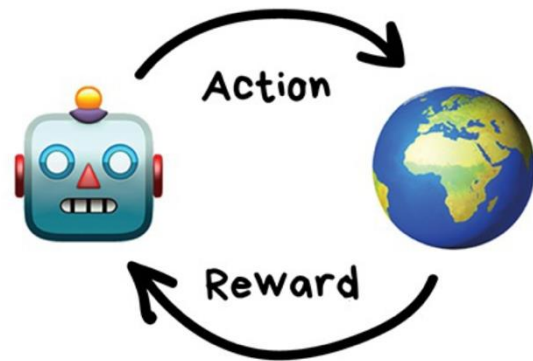
PCA: Principal
component analysis

t-SNE: t-distributed
stochastic neighbour
embedding

UMAP: Uniform manifold
approximation and
projection for dimension
reduction

Reinforcement Learning

Model that performs a series of actions by trial and error to achieve an objective



Reinforcement
Learning

Maximise reward,
e.g. reaction conditions to
optimise yield

Automated
experiments

Samples,
conditions, etc.

Agent $\rightleftharpoons$ Environment

Actions

Class Outcomes

1. Define machine learning
2. Describe the three components of machine learning with examples
3. Explain the statistical metrics used to assess model performance

Activity:

Crystal hardness
